The due date for submitting this assignment has passed.
Due on 2026-04-01, 23:59 IST.

You may submit any number of times before the due date. The final submission will be considered for grading.

You have last submitted on: 2026-04-01, 07:55 IST

Note: This assignment will be evaluated after the deadline passes. You will get your score 48 hrs after the deadline. Until then the score will be shown as Zero.

1) Consider the vector spaces V and functions $\langle ., . \rangle : V \times V \rightarrow \mathbb{R}$ defined as follows: ***1 point***

i) $V = \mathbb{R}^2$ and $\langle (x_1, x_2), (y_1, y_2) \rangle = x_1y_1 - x_2y_1 - x_1y_2 + 2x_2y_2$.
ii) $V = M_{2\times 2}(\mathbb{R})$ and $\langle A, B \rangle = Tr(AB)$,
where $Tr(M)$ denotes the trace of a matrix M, i.e., the sum of the diagonal elements of M.
iii) $V = M_{2\times 1}(\mathbb{R})$ and $\langle A, B \rangle = Tr(AB^t)$,
where $Tr(X)$ denotes the trace of a matrix X, i.e., the sum of the diagonal elements of X. Y^t denotes the transpose of matrix Y.
iv) $V = \mathbb{R}^2$ and $\langle (x_1, x_2), (y_1, y_2) \rangle = x_1y_2 + x_2y_1$.

- [x] (i) is an inner product.
- [x] (ii) is an inner product.
- [ ] (iii) is an inner product.
- [ ] (iv) is an inner product.

No, the answer is incorrect.
Score: 0

Accepted Answers:

(i) is an inner product.
(iii) is an inner product.

2) Consider two linear transformations T and S from $\mathbb{R}^2 \rightarrow \mathbb{R}^2$ defined as $T(x, y) = (2x + y, x + y)$ and $S(x, y) = (x + cy, x + 2y)$. Let A and B be matrix representations of linear transformations T and S with respect to the standard bases of $\mathbb{R}^2$ respectively. ***1 point***
Consider the following statements:

P: If $c = 1$, then A and B are similar matrices.
Q: If $c = 2$, then A and B are similar matrices.

R: If $c = 1$ and $P^{-1}AP = B$, then P can be the matrix $\begin{bmatrix} 1 & 0 \\ 1 & 2 \end{bmatrix}$.

S: If $c = 1$ and $P^{-1}AP = B$, then P can be the matrix $\begin{bmatrix} 1 & 2 \\ 1 & 1 \end{bmatrix}$.

T: If $c = 1$, then there are infinitely many P satisfying the equation $P^{-1}AP = B$.

Which of the following options are true?

- [x] P is true but Q is false.
- [ ] Both P and Q are true.
- [ ] Both R and S are true.
- [x] R is false but S is true.
- [x] T is true.

Yes, the answer is correct.

Score: 1

Accepted Answers:

P is true but Q is false.
R is false but S is true.
T is true.

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3) Let $\langle .,. \rangle$ denote the standard inner product on $\mathbb{R}^2$, i.e., $\langle (x_1, x_2), (y_1, y_2) \rangle = x_1y_1 + x_2y_2$. Which one of the following options is (are) true for the vector $\gamma \in \mathbb{R}^2$, such that $\langle \alpha, \gamma \rangle = 4$ and $\langle \beta, \gamma \rangle = 8$, where $\alpha = (3, 1)$ and $\beta = (6, 2)$. ***1 point***

- [ ] No such γ exists.
- [x] There are infinitely many such vectors which satisfy the properties of γ.
- [ ] γ is unique in $\mathbb{R}^2$.
- [x] Any vector in the set $\{(t, 4 - 3t) \mid t \in \mathbb{R}\}$ satisfies the properties of γ.
- [ ] $(1, 1)$ is the only vector which satisfies the properties of γ.

Yes, the answer is correct.

Score: 1

Accepted Answers:

There are infinitely many such vectors which satisfy the properties of y.
Any vector in the set $\{(t, 4-3t) \mid t \in \mathbb{R}\}$ satisfies the properties of y.

4) Let $\langle ., . \rangle$ denote the standard inner product on $\mathbb{R}^2$, i.e., $\langle (x_1, x_2), (y_1, y_2) \rangle = x_1y_1 + x_2y_2$ and let $v \in \mathbb{R}^2$. Consider a linear transformation $T_v : \mathbb{R}^2 \to \mathbb{R}$ defined as: ***1 point***

$$T_v(u) = \langle u, v \rangle,$$

where $v \in \mathbb{R}^2$. Which of the following options is (are) true for T_v?

- [ ] T_v is one-one for all $v = 0 \in \mathbb{R}^2$.
- [x] T_v is onto for all $v = 0 \in \mathbb{R}^2$.
- [ ] T_v is onto for all $v \in \mathbb{R}^2$.
- [x] T_v is not one-one for every $v \in \mathbb{R}^2$.
- [x] There exists a $v \in \mathbb{R}^2$ such that T_v is not onto.

Yes, the answer is correct.
Score: 1

Accepted Answers:

T_v is onto for all $v = 0 \in \mathbb{R}^2$.
T_v is not one-one for every $v \in \mathbb{R}^2$.
There exists a $v \in \mathbb{R}^2$ such that T_v is not onto.

5) Let A and B be square matrices of the same order n. Which of the following statements are sufficient to conclude that A is equivalent to B. ***1 point***

- [ ] det(A) = det(B).
- [x] Nullity(A) = Nullity(B).
- [ ] The set of columns of A, and the set of columns of B are both linearly dependent subsets of $\mathbb{R}^n$.
- [x] There exists a matrix C such that A is equivalent to C, and B is equivalent to C.
- [x] There exist invertible matrices P and Q of order n such that $PA = BQ$.

Yes, the answer is correct.

Score: 1

Accepted Answers:

Nullity(A) = Nullity(B).
There exists a matrix C such that A is equivalent to C, and B is equivalent to C.
There exist invertible matrices P and Q of order n such that $PA = BQ$.

6) Let L and L' be affine subspaces of $\mathbb{R}^3$, where $L = (0,1,1) + U$ and $L' = (0,1,0) + U'$, for some vector subspaces U and U' of $\mathbb{R}^3$. Let a basis for U be given by $\{(1,1,0),(1,0,1)\}$ and a basis for U' be given by $\{(1,0,0)\}$. Suppose there is a linear transformation $T : U \to U'$ such that $(1,0,1) \in \ker(T)$ and $T(1,1,0) = (1,0,0)$. An affine mapping $f : L \to L'$ is obtained by defining $f((0,1,1)+u) = (0,1,0) + T(u)$, for all $u \in U$. Which of the following options are true? ***1 point***

- [x] $L = \{(x, y+1, x-y+1) \mid x, y \in \mathbb{R}\}$.
- [ ] $L' = \{(x, y+1, 0) \mid x, y \in \mathbb{R}\}$.
- [ ] $L = \{(x-y, y+1, x-y+1) \mid x, y \in \mathbb{R}\}$.
- [ ] $L = \{(x, x+1, y+1) \mid x, y \in \mathbb{R}\}$.
- [x] $f(x, y+1, x-y+1) = (y, 1, 0)$
- [ ] $f(x-y, y+1, x-y+1) = (x, y+1, 0)$
- [ ] $f(x, x+1, y+1) = (y, 1, 0)$
- [ ] $f(x, y+1, x-y+1) = (0, 1, y)$

Yes, the answer is correct.

Score: 1

Accepted Answers:

$L = \{(x, y+1, x-y+1) \mid x, y \in \mathbb{R}\}$.
$f(x, y+1, x-y+1) = (y, 1, 0)$

7) Let θ be the angle between the vectors $u = (4,7,3)$ and $v = (1,2,-6)$, then what will be the value of $cos(\theta)$?

0

Yes, the answer is correct.

Score: 1

Accepted Answers:

(Type: Numeric) 0

1 point

8) Consider a basis

$$\{v_1 = (1,2,0), v_2 = (2,-1,0), v_3 = (0,0,2)\}$$

of $\mathbb{R}^3$ with usual inner product. Suppose $v = (x, y, \frac{3x+y}{5}) \in V$ is written as $v = c_1v_1 + c_2v_2 + c_3v_3$, such that $c_1 + c_2 = 4$. What will be the value of c_3?

2

Yes, the answer is correct.

Score: 1

Accepted Answers:

(Type: Numeric) 2

1 point

9) Let $V = \mathbb{R}^2$ be a vector space. Consider two inner products $\langle ., . \rangle_1$ and $\langle ., . \rangle_2$ on V defined as

$$\langle (x_1, y_1), (x_2, y_2) \rangle_1 = x_1x_2 - x_1y_2 - x_2y_1 + 4y_1y_2$$

and

$$\langle (x_1, y_1), (x_2, y_2) \rangle_2 = 3x_1x_2 + 2y_1y_2.$$

If $\langle (a, b), (4, 3) \rangle_1 = 35$ and $\langle (a, b), (4, 3) \rangle_2 = 60$, then find the value $a + 2b$.

11

Yes, the answer is correct.

Score: 1

Accepted Answers:

(Type: Numeric) 11

1 point

10) Let $V = \mathbb{R}^3$ be the inner product space with usual inner product. If θ is the angle between $(15, 8, 18)$ and (a, b, c) where $15a + 8b + 18c = 0$ and $(a^2 + b^2 + c^2) = 0$, then find the value of $\cos\theta$
.

0

Yes, the answer is correct.

Score: 1

Accepted Answers:

(Type: Numeric) 0

1 point

11) Consider a vector $v = (x - 20, 2, 1) \in \mathbb{R}^3$. Find the value of x so that the length of the vector v is minimum.

20

Yes, the answer is correct.

Score: 1

Accepted Answers:

(Type: Numeric) 20

1 point

12) Consider the following subsets of $\mathbb{R}^3$, with $\langle\ ,\ \rangle$ used to denote the standard dot product on $\mathbb{R}^3$, and $\|\ \|$ used to denote the standard length function.

$S_1 = \{v \in \mathbb{R}^3 \mid \langle v, (0, 0, 1)\rangle = 2,\ \|v\| = 1\}$

$S_2 = \{u \in \mathbb{R}^3 \mid u \in \text{Span}(\{(1, 1, 1)\}),\ \|u\| = 1\}$

If n_1 and n_2 denote the number of elements in S_1 and S_2, respectively, then find the value of $n_1 - n_2$.

-2

Yes, the answer is correct.

Score: 1

Accepted Answers:

(Type: Numeric) -2

1 point